2026 Year 12 Mathematics Standard 2 – HSC Task 1

Marking guidelines and solutions (teacher copy) • Ratios and rates & Trigonometry • 45 marks, 1 hour
NSW Mathematics Standard 11–12 Syllabus (2024), Standard 2 pathway, Year 12

Outcome and mark summary

Outcome	Section I	Section II	Total
MST-12-S2-05 solves practical problems involving ratios and rates	6	16	**22**
MST-12-S2-04 applies trigonometry to solve problems involving right-angled and non-right-angled triangles	4	19	**23**
MAO-WM-01 (working mathematically) – assessed throughout, especially the "show", "justify" and multi-step items: 14(b), 20, 21(a), 22(a), 23(a)	—		
Total	**10**	**35**	**45**

Difficulty runs from easy (core/support – nearly all students unaided) through core and medium to challenging (Band 5–6, extension-level within Standard 2). No content beyond the Standard 2 Year 12 focus areas is included.

Section I – Multiple choice (1 mark each)

Q	Ans	Working	Outcome	Level
1	**B**	1.2 L = 1200 mL; 1200 : 400 = 3 : 1	MST-12-S2-05	Easy – core/support
2	**C**	4 cm : 2 km = 4 cm : 200 000 cm = 1 : 50 000	MST-12-S2-05	Easy – core/support
3	**D**	SθW → 180° + θ = 180° + 35° = 215°	MST-12-S2-04	Easy – core/support
4	**A**	x is adjacent to 35°, 20 is the hypotenuse: x = 20 cos 35°	MST-12-S2-04	Easy – core
5	**B**	Unit prices: A \$13.60/L, B \$13.20/L, C \$13.50/L, D \$13.40/L → B lowest	MST-12-S2-05	Core
6	**D**	6/50 = 60/N → N = 60 × 50 ÷ 6 = 500	MST-12-S2-05	Core
7	**C**	Final stage: (45 − 20) km in (2.5 − 1.5) h = 25 km/h. (18 = whole-trip average; 20 = first stage)	MST-12-S2-05	Core
8	**A**	2.4 kW × 3 h = 7.2 kWh; 7.2 × \$0.30 = \$2.16	MST-12-S2-05	Core
9	**D**	Third angle = 70°; x/sin 70° = 12/sin 50° → x = 14.72… ≈ 14.7 cm	MST-12-S2-04	Medium – core
10	**C**	30 = ½ × 8 × 10 × sin θ → sin θ = 0.75 → θ = 48.59…° ≈ 49° (22° omits the ½; 131° is obtuse)	MST-12-S2-04	Challenging – Band 5

Answer spread: A ×2, B ×2, C ×3, D ×3.

Section II – Written response (35 marks)

Q	Marks	Sample answer	Criteria	Content / level / outcome
11	2	Ratio 3 : 5 : 7, total 15 parts. One part = \$1200 ÷ 15 = \$80. Cal = 7 × \$80 = **\$560**	**2** Correct answer **1** Finds total parts (15) or value of one part (\$80), or equivalent merit	Dividing in a ratio *Easy – core/support* MST-12-S2-05

12(a)	1	360° − 290° = 70° → **N70°W**	**1**	Correct answer	True → compass bearing *Easy – core/support* MST-12-S2-04
12(b)	1	052° + 180° = **232°**	**1**	Correct answer	Reverse bearing *Easy – core* MST-12-S2-04
13	2	tan 23° = h/85 → h = 85 tan 23° = 36.08… ≈ **36 m**	**2** **1**	Correct answer Correct trig equation, e.g. h = 85 tan 23°	Angle of elevation (right-angled) *Easy – core* MST-12-S2-04
14(a)	1	820 ÷ 100 × 7.4 = **60.68 L**	**1**	Correct answer	Fuel consumption *Core* MST-12-S2-05
14(b)	1	60.68 × \$2.06 = \$125.00, which is more than \$120, so **Jordan is not correct**.	**1**	Correct cost AND a conclusion consistent with it (accept follow-through from (a))	Fuel cost; justify *Core* MST-12-S2-05
15	2	2500 W = 2.5 kW. Energy = 2.5 × 6 × 90 = 1350 kWh. Cost = 1350 × \$0.32 = **\$432**	**2** **1**	Correct answer Correct energy in kWh (1350), or correct method with one error (e.g. not converting W to kW)	Energy (W → kW, kWh) and cost *Core* MST-12-S2-05
16(a)	1	14.6 × 25 000 = 365 000 cm = **3.65 km**	**1**	Correct answer	Map scale *Core* MST-12-S2-05
16(b)	2	T = D ÷ S = 3.65 ÷ 4.5 = 0.8111… h; × 60 = 48.67 min ≈ **49 minutes**	**2** **1**	Correct answer (follow-through from (a)) Correct time in hours, or correct method with incorrect conversion to minutes	Speed, distance, time *Core – medium* MST-12-S2-05
17	2	A ≈ 20/2 (32 + 41) + 20/2 (41 + 28) = 730 + 690 = **1420 m²**	**2** **1**	Correct answer One correct application of the trapezoidal rule, or equivalent merit	Trapezoidal rule (two applications) *Core – medium* MST-12-S2-05
18	2	A = 8 × 5.5 = 44 m²; 24 mm = 0.024 m; V = 44 × 0.024 = 1.056 m³; 90% → 0.9504 m³ = **950.4 L** (accept ≈ 950 L)	**2** **1**	Correct answer Correct volume of rain before 90% (1.056 m³ or 1056 L), or correct method with one error (e.g. mm not converted)	Rainfall volume, % collected *Core – medium* MST-12-S2-05
19	2	cos θ = (7² + 9² − 12²)/(2 × 7 × 9) = −14/126 = −0.1111… → θ = 96.38…° ≈ **96°**	**2** **1**	Correct answer Correct substitution into the cosine rule	Cosine rule – angle (obtuse) *Medium* MST-12-S2-04
20	3	Remaining drink: 6 − 1.5 = 4.5 L, cordial = 4.5 × 1/6 = 0.75 L, water = 3.75 L. Add 1.5 L water → water = 5.25 L. Ratio 0.75 : 5.25 = **1 : 7**	**3** **2** **1**	Correct answer Finds amounts of cordial (0.75 L) and water (5.25 L) after refilling, or equivalent merit Finds cordial and water in the remaining 4.5 L, or finds the original amounts (1 L and 5 L), or equivalent merit	Changing ratios (mixtures) *Challenging – Band 5/6* MST-12-S2-05
21(a)	1	Bearing of P from Q = 060° + 180° = 240°. ∠PQR = 240° − 150° = 90°. (Or: using the south line at Q, QP is 60° west of south (alternate angles) and QR is 180° − 150° = 30° east of south, so ∠PQR = 60° + 30° = 90°.)	**1**	Valid reasoning showing 90°	Bearings – angle in a triangle *Medium* MST-12-S2-04
21(b)	1	PR = √(18² + 25²) = √949 = 30.805… ≈ **30.8 km**	**1**	Correct answer	Pythagoras *Core* MST-12-S2-04

21(c)	2	tan ∠QPR = 25/18 → ∠QPR = 54.25…°. Bearing of R from P = 060° + 54.25° = 114.25…° ≈ **114°**	**2** **1**	Correct answer Finds ∠QPR ≈ 54°, or equivalent merit	Bearings with right-angled trig *Challenging – Band 5* MST-12-S2-04
22(a)	1	∠BOC = 245° − 130° = 115°	**1**	Correct working	Radial survey – angle from bearings *Core* MST-12-S2-04
22(b)	2	∠AOB = 130° − 25° = 105°. A = ½ × 40 × 55 × sin 105° = 1062.5… ≈ **1063 m²**	**2** **1**	Correct answer Finds ∠AOB = 105°, or correct substitution into A = ½ab sin C with an incorrect angle	Area of a triangle ½ab sin C *Medium* MST-12-S2-04
22(c)	2	$AB^2 = 40^2 + 55^2 − 2 × 40 × 55 ×$ cos 105° = 5763.8… → AB = 75.92… ≈ **75.9 m**	**2** **1**	Correct answer Correct substitution into the cosine rule	Cosine rule – side *Medium – challenging* MST-12-S2-04
23(a)	1	∠ABT = 180° − 47° = 133° (angles on a straight line). ∠ATB = 180° − 31° − 133° = 16° (angle sum of △ABT). Accept: exterior angle 47° = 31° + ∠ATB.	**1**	Valid reasoning	Angle reasoning *Medium* MST-12-S2-04
23(b)	2	BT/sin 31° = 40/sin 16° → BT = 40 sin 31° / sin 16° = 74.74… ≈ **74.7 m**	**2** **1**	Correct answer Correct sine rule statement with BT opposite 31° and 40 opposite 16°	Sine rule – side *Challenging – Band 6* MST-12-S2-04
23(c)	1	h = BT sin 47° = 74.74… × sin 47° = 54.66… ≈ **55 m** (using 74.7 gives 54.63 → 55 m)	**1**	Correct answer (follow-through from (b))	Right-angled trig *Challenging – Band 6* MST-12-S2-04

General marking notes. Accept follow-through errors in later parts where the method is correct. Accept answers that differ in the last place only because of a correct but earlier-rounded intermediate value, unless the question requires a specific rounding. Do not penalise missing units more than once across the paper.